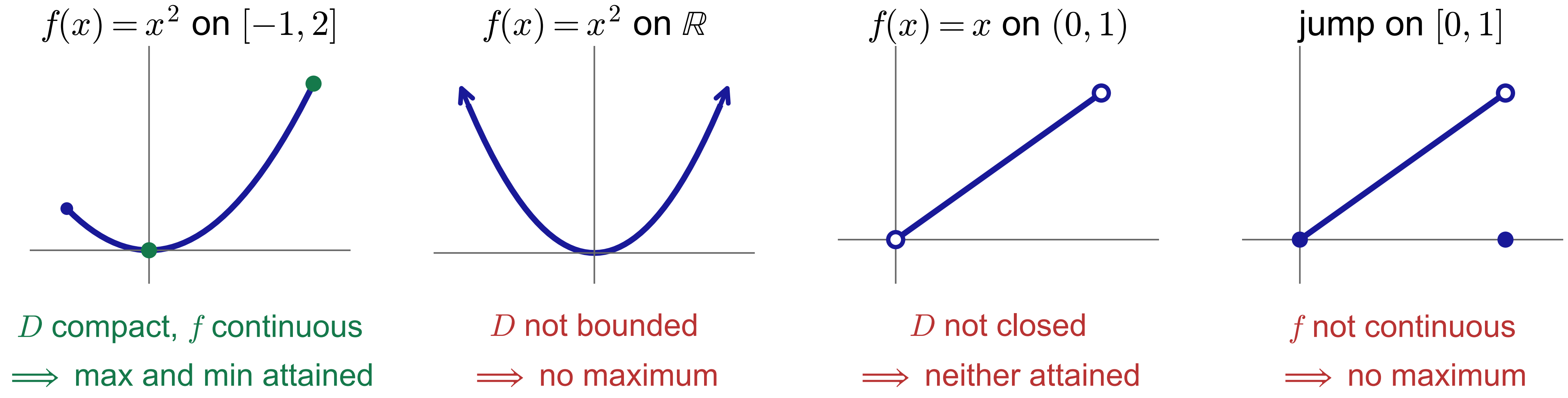

$f(x) = x^2$ on $[-1, 2]$
$f(x) = x^2$ on $\mathbb{R}$
$f(x) = x$ on $(0, 1)$
jump on $[0, 1]$
D compact, f continuous
$\implies$ max and min attained
D not bounded
$\implies$ no maximum
D not closed
$\implies$ neither attained
f not continuous
$\implies$ no maximum